Signals & Systems (151-0575-01) Exam, February 8th, 2018 - Answer key (compact)

Condensed from the official corrected solution. 50 single-choice questions, 1 point each, all-or-nothing; calculations are not graded. Each criterion heading gives the label, name and maximum, followed by the question, the options in order (A, B, C, ...) and the key. Shared context (Box) is stated once.

Context: Consider the continuous-time system dq/dt(t) = [[0, 1], [0, 0]] q(t) + [[1], [1]] u(t).

1.a - MC question 1 - 1 points:

Question: What is the system description resulting from an exact discretization with sampling time Ts?

Options: A) q[n+1] = [[0, Ts], [0, 0]] q[n] + [[Ts], [Ts]] u[n] | B) q[n+1] = [[1, Ts], [0, 1]] q[n] + [[1], [1]] u[n] | C) q[n+1] = [[1, Ts], [0, 1]] q[n] + [[Ts], [Ts]] u[n] | D) q[n+1] = [[1, Ts], [0, 1]] q[n] + [[Ts (1 + Ts/2)], [Ts]] u[n]

Key: option D - q[n+1] = [[1, Ts], [0, 1]] q[n] + [[Ts (1 + Ts/2)], [Ts]] u[n]

2.a - MC question 2 - 1 points:

Question: What is the system description resulting from a forward Euler discretization with sampling time Ts?

Options: A) q[n+1] = [[0, Ts], [0, 0]] q[n] + [[Ts], [Ts]] u[n] | B) q[n+1] = [[1, Ts], [0, 1]] q[n] + [[1], [1]] u[n] | C) q[n+1] = [[1, Ts], [0, 1]] q[n] + [[Ts], [Ts]] u[n] | D) q[n+1] = [[1, Ts], [0, 1]] q[n] + [[Ts (1 + Ts/2)], [Ts]] u[n]

Key: option C - q[n+1] = [[1, Ts], [0, 1]] q[n] + [[Ts], [Ts]] u[n]

3.a - MC question 3 - 1 points:

Question: The continuous-time signal u(t) = cos(pi/10 t) is sampled with Ts = 0.1, resulting in a discrete-time sequence {u[n]}. The signal {u[n]} is recorded for N = 1000 consecutive time steps. How many periods of {u[n]} are recorded?

Options: A) 1 | B) 2 | C) 3 | D) 4 | E) 5 | F) 6

Key: option E - 5

Context: Consider a portion of the periodic continuous-time signal x(t), plotted for t from -4 to 8 (values roughly between -0.5 and 1.5). [Figure not included.]

4.a - MC question 4 - 1 points:

Question: What is the fundamental period of the discrete-time signal {x[n]} if x(t) is sampled with Ts = 0.5?

Options: A) 1 | B) 2 | C) 4 | D) 6 | E) 8

Key: option E - 8

5.a - MC question 5 - 1 points:

Question: What is the resulting discrete-time signal {x[n]} if x(t) is sampled with Ts = 1, where the first sample is taken at t = 0?

Options: A) x[n] = 0.5 + cos(pi/4 n) - 0.3 sin(3 pi n) | B) x[n] = cos(pi/2 n) | C) x[n] = 0.5 + sin(pi/4 n - pi/2) | D) x[n] = 0.5 + cos(pi/2 n - pi/2) | E) x[n] = sin(pi/2 n - pi/2) | F) x[n] = cos(pi/4 n - pi/2) - 0.3 sin(3 pi n)

Key: option D - x[n] = 0.5 + cos(pi/2 n - pi/2)

Context: Consider the continuous-time signal x(t) = cos(6pi/5 t) + cos(2 pi t) and the discrete-time signals x_1[n] = 1 + cos(4pi/5 n), x_2[n] = 1 + cos(6pi/5 n), x_3[n] = cos(6pi/5 n + 3pi/10).

6.a - MC question 6 - 1 points:

Question: Which discrete-time sequences result from sampling x(t) with sampling time Ts = 1, where the first sample is taken at t_0 = 0?

Options: A) None | B) x_1[n] | C) x_2[n] | D) x_3[n] | E) x_1[n] or x_2[n] | F) x_1[n] or x_3[n] | G) x_2[n] or x_3[n] | H) x_1[n], x_2[n], or x_3[n]

Key: option E - x_1[n] or x_2[n]

7.a - MC question 7 - 1 points:

Question: Which discrete-time sequences result from sampling x(t) with sampling time Ts = 1, where the first sample is taken at t_0 = 0.25?

Options: A) None | B) x_1[n] | C) x_2[n] | D) x_3[n] | E) x_1[n] or x_2[n] | F) x_1[n] or x_3[n] | G) x_2[n] or x_3[n] | H) x_1[n], x_2[n], or x_3[n]

Key: option D - x_3[n]

Context: Consider the following systems:
G_1: y[n] = u[|2n|]
G_2: y[n] = u[n] + u[n+1]
G_3: y[n] = 1 y[n-1] - 1/4 y[n-2] + u[n], causal
G_4: y[n] = 1/2 (y[n-1])^2 + u[n], causal

8.a - MC question 8 - 1 points:

Question: Select all linear systems.

Options: A) G_1, G_2 | B) G_1, G_3 | C) G_1, G_4 | D) G_2, G_3 | E) G_2, G_4 | F) G_3, G_4 | G) G_1, G_2, G_3 | H) G_2, G_3, G_4

Key: option G - G_1, G_2, G_3

9.a - MC question 9 - 1 points:

Question: Select all time-invariant systems.

Options: A) G_1, G_2 | B) G_1, G_3 | C) G_1, G_4 | D) G_2, G_3 | E) G_2, G_4 | F) G_3, G_4 | G) G_1, G_2, G_3 | H) G_2, G_3, G_4

Key: option H - G_2, G_3, G_4

10.a - MC question 10 - 1 points:

Question: Select all bounded-input bounded-output stable systems.

Options: A) G_1, G_2 | B) G_1, G_3 | C) G_1, G_4 | D) G_2, G_3 | E) G_2, G_4 | F) G_3, G_4 | G) G_1, G_2, G_3 | H) G_2, G_3, G_4

Key: option G - G_1, G_2, G_3

Context: Consider the following representations of systems:
G_1: h = {..., 0, 1 (n=0), 2, 3, 1, 0, ...}
G_2: y[n] = 2 y[n-1] + u[n], anti-causal
G_3: H(z) = z / (z^2 - 0.5 z + 0.25)

11.a - MC question 11 - 1 points:

Question: Which systems are guaranteed to be stable?

Options: A) G_1 | B) G_2 | C) G_3 | D) G_1, G_2 | E) G_1, G_3 | F) G_2, G_3 | G) G_1, G_2, G_3

Key: option D - G_1, G_2

Context: Consider portions (n = -1..7) of the responses {y_1[n]} and {y_2[n]} of two systems to the unit step input {s[n]} (s[n] = 1 if n >= 0, 0 otherwise). [Figure not included.]

12.a - MC question 12 - 1 points:

Question: Which of the system responses could result from an FIR filter?

Options: A) {y_1[n]} | B) {y_2[n]} | C) None of them | D) {y_1[n]} and {y_2[n]}

Key: option D - {y_1[n]} and {y_2[n]}

13.a - MC question 13 - 1 points:

Question: Consider the system response {y_1[n]}. Which difference equation could represent the system?

Options: A) y[n] = 1/2 y[n-1] + u[n] | B) y[n] = 3/2 u[n] + u[n-1] + u[n-2] | C) y[n] = -1/2 y[n-1] + 2 u[n] | D) y[n] = 2 u[n] - 1/2 u[n-1] | E) y[n] = 2/3 y[n-1] + u[n-1]

Key: option E - y[n] = 2/3 y[n-1] + u[n-1]

14.a - MC question 14 - 1 points:

Question: Consider the system response {y_2[n]}. Which difference equation could represent the system?

Options: A) y[n] = 2/3 y[n] + u[n-1] + u[n-2] | B) y[n] = 1/2 y[n-1] - 1/4 y[n-2] + u[n] | C) y[n] = u[n] + 3/4 u[n-1] + 1/4 u[n-2] | D) y[n] = u[n] + 1/2 u[n-1] + 1/2 u[n-2] + 1/2 u[n-3] | E) y[n] = u[n] + 1/2 u[n-1] + 1/2 u[n-2]

Key: option C - y[n] = u[n] + 3/4 u[n-1] + 1/4 u[n-2]

Context: The impulse responses h_1, h_2, h_3, h_4 (n = 0..3) of four causal FIR filters are convolved with a sequence u (n = 0..50, values in {-1, 0, 1}), resulting in four output sequences y_A, y_B, y_C, y_D. [Figure not included.]

15.a - MC question 15 - 1 points:

Question: Which output sequence y corresponds to y = h_3 * u?

Options: A) y = y_A | B) y = y_B | C) y = y_C | D) y = y_D

Key: option A - y = y_A

16.a - MC question 16 - 1 points:

Question: Which impulse response h corresponds to y_B = h * u?

Options: A) h = h_1 | B) h = h_2 | C) h = h_3 | D) h = h_4

Key: option D - h = h_4

Context: Consider the following state-space descriptions of causal, LTI DT systems, q[n+1] = A q[n] + B u[n], y[n] = C q[n] + D u[n]:
G_1: A = [[0, 1, 1], [0, 0, 1/2], [0, 0, 0]], B = [[1], [1], [-1]], C = [0, 0, 1], D = 0
G_2: A = 1, B = 1, C = 1, D = 0
G_3: A = [[0, 1/2], [1, 0]], B = [[1], [0]], C = [1, 1/2], D = 1
G_4: A = [[1/2, 0], [0, 0]], B = [[1], [1]], C = [1, 0], D = 0
G_5: A = 0, B = 1, C = 1, D = 1

17.a - MC question 17 - 1 points:

Question: Select the answer containing all stable systems.

Options: A) G_1, G_2 | B) G_1, G_2, G_5 | C) G_5 | D) G_1, G_3, G_4, G_5 | E) G_1 | F) G_4 | G) G_1, G_5 | H) G_3

Key: option D - G_1, G_3, G_4, G_5

18.a - MC question 18 - 1 points:

Question: Select the answer containing all systems with an FIR.

Options: A) G_1 | B) G_2 | C) G_3 | D) G_4 | E) G_5 | F) G_1, G_2 | G) G_2, G_3 | H) G_4, G_5 | I) G_1, G_5

Key: option I - G_1, G_5

19.a - MC question 19 - 1 points:

Question: Which is a state-space representation of a system with LCCDE y[n] = 1/2 y[n-2] + u[n] + u[n-1]?

Options: A) G_1 | B) G_2 | C) G_3 | D) G_4 | E) G_5

Key: option C - G_3

Context: Consider portions (n = -4..4) of the impulse responses {h_i[n]} of stable LTI systems G_i, i in {1, ..., 6}. [Figure not included.]

20.a - MC question 20 - 1 points:

Question: Which system has transfer function H(z) = z + (1/2 z) / (z - 1/2)?

Options: A) G_1 | B) G_2 | C) G_3 | D) G_4 | E) G_5 | F) G_6

Key: option B - G_2

Context: Consider a causal system with the difference equation y[n] = y[n-1] - 1/4 y[n-2] + u[n-1].

21.a - MC question 21 - 1 points:

Question: The input u[n] = 1 if n >= 0, 0 otherwise, is applied to the system. What is the value of y[n] for n -> infinity?

Options: A) 3 | B) 3.5 | C) 4 | D) 4.5 | E) 5

Key: option C - 4

22.a - MC question 22 - 1 points:

Question: A periodic signal u[n] = 1/2 cos(pi n) is applied to the system. What is the output of the system?

Options: A) y[n] = 2 cos(pi n - pi) | B) y[n] = 2/3 cos(pi n - pi) | C) y[n] = 2/9 cos(pi n - pi) | D) y[n] = 2 cos(pi n) | E) y[n] = 2/3 cos(pi n) | F) y[n] = 2/9 cos(pi n)

Key: option C - y[n] = 2/9 cos(pi n - pi)

Context: Consider a causal LTI system G_* (y_* = G_* u_*) given by its magnitude and phase frequency response plots for Omega in [0, pi]. [Figure not included.] G_* is used to create two systems: G_A is the series connection G_* -> G_*; G_B is the parallel connection of two copies of G_*, whose outputs are summed. The input to both G_A and G_B is u[n] = 2 cos(pi/2 n) + 3/2 cos(2pi/3 n) + sin(pi n).

23.a - MC question 23 - 1 points:

Question: What is the resulting output y = G_A u?

Options: A) y[n] = 2/9 cos(pi/2 n) + 1/9 sin(pi n) | B) y[n] = -4/3 cos(pi/2 n) + cos(2pi/3 n) + 2/3 sin(pi n) | C) y[n] = 4/3 sin(pi/2 n - pi/2) + 2/3 sin(pi n) | D) y[n] = 2/3 cos(pi/2 n) + 3/2 cos(2pi/3 n) + 1/3 sin(pi n) | E) y[n] = -2/9 cos(pi/2 n) + 1/9 sin(pi n) | F) y[n] = 2/3 sin(pi/2 n) + 1/3 sin(pi n)

Key: option E - y[n] = -2/9 cos(pi/2 n) + 1/9 sin(pi n)

24.a - MC question 24 - 1 points:

Question: What is the resulting output y = G_B u?

Options: A) y[n] = -4/3 cos(pi/2 n) + cos(2pi/3 n) + 2/3 sin(pi n) | B) y[n] = 2/3 sin(pi/2 n) + 1/3 sin(pi n) | C) y[n] = -2/9 cos(pi/2 n) + 1/9 sin(pi n) | D) y[n] = 4/3 cos(pi/2 n - pi/2) + 2/3 sin(pi n) | E) y[n] = 2/9 cos(pi/2 n) + 1/9 sin(pi n) | F) y[n] = 2/3 cos(pi/2 n) + 3/2 cos(2pi/3 n) + 1/3 sin(pi n)

Key: option D - y[n] = 4/3 cos(pi/2 n - pi/2) + 2/3 sin(pi n)

Context: Consider DT Fourier magnitude plots of four different causal sequences: X_A(Omega), X_B(Omega), X_C(Omega), X_D(Omega), for Omega in [0, pi]. [Figure not included.]

25.a - MC question 25 - 1 points:

Question: Which could be the magnitude plot corresponding to a given causal sequence x[n] (stem plot for n = 0..10)? [Figure not included.]

Options: A) X_A(Omega) | B) X_B(Omega) | C) X_C(Omega) | D) X_D(Omega)

Key: option A - X_A(Omega)

26.a - MC question 26 - 1 points:

Question: Which could be the magnitude plot corresponding to another given causal sequence x[n] (stem plot for n = 0..10)? [Figure not included.]

Options: A) X_A(Omega) | B) X_B(Omega) | C) X_C(Omega) | D) X_D(Omega)

Key: option B - X_B(Omega)

Context: Consider the magnitude and phase response of a stable LTI system G, for Omega in [0, pi]. [Figure not included.]

27.a - MC question 27 - 1 points:

Question: Which is the transfer function of the LTI system G?

Options: A) H(z) = 0.5 (z + 1) / (z - 0.5) | B) H(z) = 0.35 (z + 1) / (z - 0.3) | C) H(z) = (z + 1)^2 / (4 z^2) | D) H(z) = (z + 1) / (2z) | E) H(z) = (z + 0.73) / (z^2 - 0.3)

Key: option B - H(z) = 0.35 (z + 1) / (z - 0.3)

28.a - MC question 28 - 1 points:

Question: The LTI system is cascaded with a delay element G_D with transfer function H_D(z) = z^-1 (u -> G -> G_D -> y). Which of four magnitude/phase plots (Plot A to Plot D) shows the response of the resulting system G_casc? [Figure not included.]

Options: A) Plot A | B) Plot B | C) Plot C | D) Plot D

Key: option D - Plot D

29.a - MC question 29 - 1 points:

Question: The sequence {x[n]} = k * {-1 (n=0), -2, -1, 1, 2, 1} is periodic with period N = 6, where k is an integer. The corresponding DFS coefficients are {X[k]} = {0 (k=0), X[1], 0, 0, 0, X[5]}. What is the magnitude of coefficient X[1]?

Options: A) |X[1]| = 3|k|^2 | B) |X[1]| = sqrt(72)|k| | C) |X[1]| = 6|k| | D) |X[1]| = 36|k| | E) |X[1]| = 36|k|^2 | F) |X[1]| = 3|k|

Key: option C - |X[1]| = 6|k|

Context: Consider the DT signal x[n] = 2 e^(j 4pi/7 n) + 2 e^(j 10pi/7 n) for all times n. A DFT of the first N = 21 samples of {x[n]} is taken. Let {X[k]} be the corresponding DFT coefficients.

30.a - MC question 30 - 1 points:

Question: What is the value of X[1]?

Options: A) X[1] = 2/3 | B) X[1] = 1/3 | C) X[1] = 42 | D) X[1] = 2 | E) X[1] = 6 | F) X[1] = 0

Key: option F - X[1] = 0

31.a - MC question 31 - 1 points:

Question: What is the value of X[6]?

Options: A) X[6] = 2 | B) X[6] = 6 | C) X[6] = 42 | D) X[6] = 0 | E) X[6] = 2/3 | F) X[6] = 1/3

Key: option C - X[6] = 42

32.a - MC question 32 - 1 points:

Question: The magnitudes of the DFT coefficients computed with the first N = 4 samples of a real discrete-time signal {x[n]} are |X[k]| = {0, 2, 0, 2} for k = 0..3. Which of four plots (Plot A to Plot D) could show the magnitudes of the DFT coefficients computed with the first N = 6 samples of the same signal? [Figure not included.]

Options: A) Plot A | B) Plot B | C) Plot C | D) Plot D

Key: option C - Plot C

33.a - MC question 33 - 1 points:

Question: Consider the transfer function of system G: H(z) = (z^2 + 1/2 z) / (z^2 - 1/4). What is the impulse response of G?

Options: A) h = {..., 0, 2 (n=0), 1, 1, 1, 0, ...} | B) h = {..., 0, 2 (n=0), 1, 1/2, 1/4, 0, ...} | C) h = {..., 0, 1 (n=0), 1/3, 1/9, 1/27, ...} | D) h = {..., 0, 1 (n=0), 1/2, 1/4, 1/8, ...} | E) h = {..., 0, 1, 1 (n=0), 1/2, 1/4, 0, ...} | F) h = {..., 3, 1 (n=0), 1/3, 1/9, 1/27, ...}

Key: option D - h = {..., 0, 1 (n=0), 1/2, 1/4, 1/8, ...}

34.a - MC question 34 - 1 points:

Question: Which of six magnitude responses (Plot A to Plot F) belongs to a system with transfer function H(z) = 1/8 (z - 1) / (z^2 + z + 0.5)? [Figure not included.]

Options: A) Plot A | B) Plot B | C) Plot C | D) Plot D | E) Plot E | F) Plot F

Key: option A - Plot A

35.a - MC question 35 - 1 points:

Question: Let G be a causal LTI system whose pole-zero plot is given (poles x, zeros o, unit circle shown). [Figure not included.] Which of four magnitude responses (Plot A to Plot D) could belong to system G? [Figure not included.]

Options: A) Plot A | B) Plot B | C) Plot C | D) Plot D

Key: option A - Plot A

36.a - MC question 36 - 1 points:

Question: Consider a CT low-pass filter H_LP(s) and a CT high-pass filter H_HP(s), given by their Bode magnitude plots for omega from 10^-3 to 10^3 rad/s. [Figure not included.] Which frequency transformation was used to obtain H_HP(s) from H_LP(s)?

Options: A) H_HP(s) = H_LP(1/(10s)) | B) H_HP(s) = H_LP(s^2) | C) H_HP(s) = H_LP((s^2 + 1)/s) | D) H_HP(s) = H_LP(s/10) | E) H_HP(s) = H_LP(10/s) | F) H_HP(s) = H_LP(10s)

Key: option A - H_HP(s) = H_LP(1/(10s))

Context: Consider the system G built as follows: the input u goes through G_1 followed by G_2 (series path), and in parallel through G_3; the outputs of the two paths are summed to give y. Available systems for G_i are gains H_i(z) = a, low-pass filters H_i(z) = H_LP(z), and high-pass filters H_i(z) = H_HP(z). Note that you only have to choose the class of system, not its specifications.

37.a - MC question 37 - 1 points:

Question: How can G_1, G_2 and G_3 be chosen to design G to be a band-stop filter?

Options: A) G_1: H_1(z) = H_LP(z); G_2: H_2(z) = H_HP(z); G_3: H_3(z) = 0 | B) G_1: H_1(z) = H_LP(z); G_2: H_2(z) = 1; G_3: H_3(z) = H_LP(z) | C) G_1: H_1(z) = H_LP(z); G_2: H_2(z) = 1; G_3: H_3(z) = H_HP(z) | D) G_1: H_1(z) = H_LP(z); G_2: H_2(z) = H_HP(z); G_3: H_3(z) = 1 | E) G_1: H_1(z) = H_HP(z); G_2: H_2(z) = 0; G_3: H_3(z) = H_LP(z) | F) G_1: H_1(z) = H_LP(z); G_2: H_2(z) = 0; G_3: H_3(z) = H_HP(z)

Key: option C - G_1: H_1(z) = H_LP(z); G_2: H_2(z) = 1; G_3: H_3(z) = H_HP(z)

38.a - MC question 38 - 1 points:

Question: How can G_1, G_2 and G_3 be chosen to design G to be a band-pass filter?

Options: A) G_1: H_1(z) = H_LP(z); G_2: H_2(z) = 0; G_3: H_3(z) = H_HP(z) | B) G_1: H_1(z) = H_LP(z); G_2: H_2(z) = 1; G_3: H_3(z) = H_HP(z) | C) G_1: H_1(z) = H_LP(z); G_2: H_2(z) = H_HP(z); G_3: H_3(z) = 1 | D) G_1: H_1(z) = H_LP(z); G_2: H_2(z) = 1; G_3: H_3(z) = H_LP(z) | E) G_1: H_1(z) = H_LP(z); G_2: H_2(z) = H_HP(z); G_3: H_3(z) = 0 | F) G_1: H_1(z) = H_HP(z); G_2: H_2(z) = 0; G_3: H_3(z) = H_LP(z)

Key: option E - G_1: H_1(z) = H_LP(z); G_2: H_2(z) = H_HP(z); G_3: H_3(z) = 0

Context: A second-order IIR filter with transfer function H(z) = (b_0 + b_1 z^-1 + b_2 z^-2) / (1 + a_1 z^-1 + a_2 z^-2) (1) is realized with the following block diagram. On the left, u passes through two cascaded delay elements G_d, giving u[n-1] and u[n-2]; u[n], u[n-1], u[n-2] are scaled by gains g_0, g_1, g_2. On the right, y passes through two cascaded delay elements G_d, giving y[n-1] and y[n-2], scaled by gains g_4 and g_5. A bottom adder sums g_2 u[n-2] and g_5 y[n-2]; a middle adder sums g_1 u[n-1], g_4 y[n-1] and the bottom adder output; a top adder sums g_0 u[n] and the middle adder output. The top adder output is scaled by g_3 to give y. The delay elements have transfer function H_d(z) = z^-1, and a gain g maps x_1 to x_2 = g x_1, g in R.

39.a - MC question 39 - 1 points:

Question: How should the gains be chosen such that the second-order IIR filter (1) is realized?

Options: A) g_0 = 1, g_1 = a_1, g_2 = a_2, g_3 = b_0, g_4 = b_1, g_5 = b_2 | B) g_0 = 1, g_1 = a_1, g_2 = a_2, g_3 = b_0, g_4 = -b_1, g_5 = -b_2 | C) g_0 = b_0, g_1 = b_1, g_2 = b_2, g_3 = 1, g_4 = a_1, g_5 = a_2 | D) g_0 = b_0, g_1 = b_1, g_2 = b_2, g_3 = 1, g_4 = -a_1, g_5 = -a_2

Key: option D - g_0 = b_0, g_1 = b_1, g_2 = b_2, g_3 = 1, g_4 = -a_1, g_5 = -a_2

40.a - MC question 40 - 1 points:

Question: The form above needs four delay units to realize a second-order IIR filter with transfer function (1). Which statement is correct?

Options: A) Five is the minimum number of delay elements needed to realize a general second order IIR filter. Four elements were only sufficient above because of a_0 = 1 in (1). | B) Four is the minimum number of delay elements needed to realize a general second order IIR filter. | C) Three is the minimum number of delay elements needed to realize a general second order IIR filter. | D) Two is the minimum number of delay elements needed to realize a general second order IIR filter. | E) One is the minimum number of delay elements needed to realize a general second order IIR filter.

Key: option D - Two is the minimum number of delay elements needed to realize a general second order IIR filter.

Context: Consider a sequence {u[n]}, n = 0..100, filtered by four different causal discrete-time filters G_i, i in {1, 2, 3, 4}, resulting in the sequences y_i = G_i u. [Figure not included.]

41.a - MC question 41 - 1 points:

Question: Which filter could have the given magnitude response (close to 0 at Omega = 0, rising to about 1 by Omega = pi/4 and staying near 1 up to pi)?

Options: A) G_2 | B) G_4 | C) G_1 | D) G_3

Key: option B - G_4

42.a - MC question 42 - 1 points:

Question: Which could be the gain of the filter G_3 at Omega = pi/2?

Options: A) |H_3(Omega = pi/2)| = 0 | B) |H_3(Omega = pi/2)| = 2 | C) |H_3(Omega = pi/2)| = 4 | D) |H_3(Omega = pi/2)| = 0.98 | E) |H_3(Omega = pi/2)| = pi

Key: option A - |H_3(Omega = pi/2)| = 0

43.a - MC question 43 - 1 points:

Question: Select the answer containing all filter(s) which could have an FIR shorter than 4 samples.

Options: A) G_1 | B) G_2 | C) G_3 | D) G_4 | E) G_1, G_2 | F) G_1, G_3 | G) G_1, G_4 | H) G_2, G_3 | I) G_2, G_4

Key: option F - G_1, G_3

44.a - MC question 44 - 1 points:

Question: Which filter could be a median filter?

Options: A) G_1 | B) G_2 | C) G_3 | D) G_4

Key: option A - G_1

45.a - MC question 45 - 1 points:

Question: Consider a continuous-time, causal system with transfer function H(s) = (beta s + 1) / (alpha s^2 + s + 1) with alpha > 0 and beta > 0. The system is discretized using the bilinear transform and sampling time Ts. What is the phase of the discrete-time transfer function, angle H(z), at z = e^(j pi)?

Options: A) angle H(z) = -pi/2 | B) angle H(z) = -pi/4 | C) angle H(z) = 0 | D) angle H(z) = pi/2 | E) angle H(z) = pi/4 | F) angle H(z) = pi

Key: option A - angle H(z) = -pi/2

Context: Consider the following system descriptions:
G_1: h = {..., 0, 1 (n=0), 1/2, 1/3, 0, ...}
G_2: y[n] = 1/2 y[n-1] + u[n]
G_3: y[n] = u[n-1]
G_4: y[n] = u[n+1] + 2 u[n] + u[n-1]

46.a - MC question 46 - 1 points:

Question: Select all systems whose phase response is zero.

Options: A) G_1 | B) G_2 | C) G_3 | D) G_4 | E) G_1, G_2 | F) G_1, G_3 | G) G_1, G_4 | H) G_2, G_3 | I) G_2, G_4 | J) G_3, G_4

Key: option D - G_4

Context: The objective of the following MATLAB code is to estimate the transfer function coefficients a_1, b_0 and b_1 of H(z) = (b_0 + b_1 z^-1) / (1 + a_1 z^-1), given estimated frequency responses H_hat(Omega) for frequencies INCLUDING Omega = 0 and Omega = pi, saved in MATLAB vectors H.mat and Omega.mat, respectively.

```
1  %% First order transfer function estimation
2
3  % load estimated frequency responses at tested frequencies
4  load('H.mat'); % H is a 1xL matrix with the estimated freq resp
5  load('Omega.mat'); % Omega is a 1xL matrix with the corresponding freq
6
7  % number of frequencies tested
8  L=length(Omega);
9
10 % assert that enough frequencies were tested
11 assert(...); % assert(cond) throws an error if cond is false.
12
13 % matrices for least-squares problem
14 F = zeros(2*L, 3);
15 G = zeros(2*L, 1);
16
17 for l=0:L-1
18     R = abs(H(l+1)); % the magnitude of the estimated freq resp
19     Phi = phase(H(l+1)); % the phase of the estimated freq resp
20     Om = Omega(l+1); % the freq corresponding to the estimated freq resp
21
22     % populate F and G
23     F(2*l+1,:) = ...;
24     F(2*l+2,:) = [-R*sin(Phi-Om), 0, -sin(Om)];
25     G(2*l+1)   = R*cos(Phi);
26     G(2*l+2)   = R*sin(Phi);
27 end
28
29 % now estimate the transfer function coefficients using least-squares
30 Theta = F\G;
31
32 % define the transfer function;
33 % remember, a0 = 1 is not in the Theta parameter vector
34 a = [1; Theta(1)].';
35 b = Theta(2:3).';
```

47.a - MC question 47 - 1 points:

Question: Which option correctly completes line 11?

Options: A) assert(2*L>6); | B) assert(2*L>5); | C) assert(2*L>3); | D) assert(2*L>2); | E) assert(2*L>1);

Key: option B - assert(2*L>5);

48.a - MC question 48 - 1 points:

Question: Which option correctly completes line 23?

Options: A) F(2*l+1,:) = [-R*cos(Om), 1, cos(Phi-Om)]; | B) F(2*l+1,:) = [-R*cos(Phi-Om), 1, cos(Phi-Om)]; | C) F(2*l+1,:) = [R*cos(Phi-Om), 1, cos(Om)]; | D) F(2*l+1,:) = [-R*cos(Phi-Om), 0, cos(Om)]; | E) F(2*l+1,:) = [-R*cos(Phi-Om), 1, cos(Om)];

Key: option E - F(2*l+1,:) = [-R*cos(Phi-Om), 1, cos(Om)];

49.a - MC question 49 - 1 points:

Question: To determine the frequency response H(Omega) of a stable LTI system at frequency Omega = pi/4, a sinusoidal input {u_e[n]} = {A cos(pi/4 n)} is applied and the output {y_m[n]} is measured. The first N_T samples are discarded and the following N samples are used to calculate the DFT coefficients Y_m[l] = sum_{n=N_T}^{N_T+N-1} y_m[n] e^(-j 2pi/N l n) and U_e[l] = sum_{n=N_T}^{N_T+N-1} u_e[n] e^(-j 2pi/N l n), where N_T = 200 and N = 1000. Which is the frequency response estimate at Omega = pi/4?

Options: A) H_hat(pi/4) = Y_m[150] / U_e[150] | B) H_hat(pi/4) = Y_m[25] / U_e[25] | C) H_hat(pi/4) = Y_m[250] / U_e[250] | D) H_hat(pi/4) = Y_m[4] / U_e[4] | E) H_hat(pi/4) = Y_m[50] / U_e[50] | F) H_hat(pi/4) = Y_m[125] / U_e[125]

Key: option F - H_hat(pi/4) = Y_m[125] / U_e[125]

50.a - MC question 50 - 1 points:

Question: You conducted three experiments to identify an unknown transfer function and obtained the magnitude response estimates |H_hat(Omega = 0)| = 0.05, |H_hat(Omega = pi/2)| = 0.39, |H_hat(Omega = pi)| = 1.96. Which of the following transfer functions fits your experiments best?

Options: A) H(z) = 1/2 (z + 1) / (z - 0.5) | B) H(z) = 1/2 (z^2 - 0.5) / (z^2 + 0.5) | C) H(z) = 1/3 z / (z + 0.75) | D) H(z) = 1/3 (z - 1) / (z + 2/3)

Key: option D - H(z) = 1/3 (z - 1) / (z + 2/3)